# IEOR E4004: Project 2

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https://github.com/angelhuang1108/Scheduling-the-NBA

## Introduction

Scheduling a professional sports league is a hard combinatorial problem with real consequences for travel cost, player health, and competitive fairness. The NBA's actual schedule has to balance arena availability, broadcast windows, back-to-back limits, and total miles flown across 30 teams over 82 games each. The league has used optimization software to build its schedule for over a decade, and small improvements in mileage or rest distribution translate to millions of dollars and measurable performance differences (Entine and Small 2008).

This project takes a smaller version of that problem: a fictitious 16-team, 16-date NBA schedule, given as a draft in *games.csv*. Our goal is to model the schedule as an integer program and use it to (1) summarize the schedule's structure team-by-team, (2) write an IP whose feasible solutions preserve that structure but allow opponents to be reassigned, (3) add a time zone constraint limiting how much travel disorientation any team can be forced into over three consecutive games, and (4) find concrete improvements to the original schedule. The rest of the report walks through the parameters we estimated from the data, the four IP formulations, the results from each, and what they reveal about the tradeoff between league-wide efficiency and team-by-team fairness.

## Parameter Estimation

Most of our model parameters come directly from *games.csv*. The set of teams T, the set of dates D, the home and away date sets H(i) and A(i), and the home and away matchup counts $c^H(i,j)$ and $c^A(i,j)$ are all extracted by parsing the file, as detailed in Question 1. Two additional parameters needed for Questions 3 and 4 (arena time zones and arena coordinates) are not in the file and were collected separately.

For Question 3 we assigned each of the 16 arenas to one of four time zones (Eastern (UTC−5), Central (UTC−6), Mountain (UTC−7), or Pacific (UTC−8)) based on the arena's city. We use static UTC offsets and ignore daylight saving transitions. This is a deliberate simplification, but since all 16 game dates fall in November and December the affected window is short and the offsets are stable for most of the schedule.

For Question 4 we collected (latitude, longitude) for each arena from Google Maps, rounded to four decimal places (an accuracy of about 10 meters, which is well below the relevant scale). Distances between any two arenas are then computed using the Haversine formula:

$$d = 2R \cdot \arcsin\left(\sqrt{sin^2\left(\frac{\varphi_2-\varphi_1}{2}\right) + \cos(\varphi_1)\cdot\cos(\varphi_2)\cdot sin^2\left(\frac{\lambda_2-\lambda_1}{2}\right)}\right)$$

$\varphi_i$ and $\lambda_i$ are the latitude and longitude of arena *i* in radians and $R = 3958.8$ miles is the radius of the Earth. The Haversine distance is the great-circle distance and is the natural approximation for flight mileage at this scale, actual charter flight paths add a few percent of overhead, which is small enough to ignore for the comparisons we make. The full distance matrix is precomputed once and treated as a constant in the IP.

# Question 1

For each team *i* we found the following information from the preliminary schedule:

- (a) All dates when team *i* played at home
- (b) For each opponent *j*, the number of times team *i* hosted *j* at home
- (c) For each opponent *j*, the number of times team *i* played away at *j*'s arena
- (d) All dates when team *i* played away

## (a) & (d) Home and Away Dates

The table below lists each team's home game dates (a) and away game dates (d). The schedule spans November 1 - December 25, 2025, across 16 dates, with 8 games per date (128 total).

| Team | (a) Home Dates | (d) Away Dates |
|---|---|---|
| **Atlanta Hawks** | Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Nov 29, Dec 25 | Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Dec 01 |
| **Boston Celtics** | Nov 01, Nov 07, Nov 13, Nov 19, Nov 23, Nov 28 | Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 21, Nov 27, Nov 29, Dec 01, Dec 25 |
| **Brooklyn Nets** | Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Nov 29, Dec 01 | Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25 |
| **Chicago Bulls** | Nov 01, Nov 03, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01 | Nov 05, Nov 07, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25 |
| **Cleveland Cavaliers** | Nov 03, Nov 07, Nov 17, Nov 19, Nov 21, Nov 23, Nov 28, Nov 29, Dec 01, Dec 25 | Nov 01, Nov 05, Nov 11, Nov 13, Nov 15, Nov 27 |
| **Dallas Mavericks** | Nov 03, Nov 07, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25 | Nov 01, Nov 05, Nov 11, Nov 13, Nov 21, Nov 29, Dec 01 |
| **Denver Nuggets** | Nov 05, Nov 07, Nov 11, Nov 15, Nov 17, Nov 19, Nov 21, Dec 25 | Nov 01, Nov 03, Nov 13, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01 |
| **Golden State Warriors** | Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 27, Dec 25 | Nov 01, Nov 07, Nov 13, Nov 19, Nov 21, Nov 23, Nov 28, Nov 29, Dec 01 |
| **Houston Rockets** | Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Nov 27, Dec 25 | Nov 01, Nov 05, Nov 07, Nov 17, Nov 28, Nov 29, Dec 01 |
| **Los Angeles Lakers** | Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25 | Nov 01, Nov 07, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01 |
| **Miami Heat** | Nov 01, Nov 05, Nov 13, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01 | Nov 03, Nov 07, Nov 11, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25 |
| **Milwaukee Bucks** | Nov 01, Nov 07, Nov 13, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 01 | Nov 03, Nov 05, Nov 11, Nov 15, Nov 17, Nov 19, Dec 25 |
| **New York Knicks** | Nov 01, Nov 05, Nov 11, Nov 13, Nov 15, Dec 01 | Nov 03, Nov 07, Nov 17, Nov 19, Nov 21, Nov 23, Nov 27, Nov 28, Nov 29, Dec 25 |
| **Philadelphia 76ers** | Nov 01, Nov 05, Nov 07, Nov 11, Nov 21, Nov 29, Dec 01 | Nov 03, Nov 13, Nov 15, Nov 17, Nov 19, Nov 23, Nov 27, Nov 28, Dec 25 |
| **Phoenix Suns** | Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25 | Nov 01, Nov 05, Nov 07, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01 |
| **Toronto Raptors** | Nov 01, Nov 05, Nov 07, Nov 17, Nov 27, Nov 28, Nov 29, Dec 01 | Nov 03, Nov 11, Nov 13, Nov 15, Nov 19, Nov 21, Nov 23, Dec 25 |

## (b) Home Matchup Counts: $c^H(i, j)$

Each cell shows the number of times the row team (*i*) hosted the column team (*j*) at home.

| Team \ Opp. | ATL | BOS | BKN | CHI | CLE | DAL | DEN | GSW | HOU | LAL | MIA | MIL | NYK | PHI | PHX | TOR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Atlanta Hawks** | - | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |

| Team \ Opp. | ATL | BOS | BKN | CHI | CLE | DAL | DEN | GSW | HOU | LAL | MIA | MIL | NYK | PHI | PHX | TOR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Boston Celtics** | 0 | - | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |
| **Brooklyn Nets** | 1 | 1 | - | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| **Chicago Bulls** | 1 | 1 | 1 | - | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| **Cleveland Cavaliers** | 1 | 1 | 1 | 1 | - | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| **Dallas Mavericks** | 0 | 0 | 0 | 1 | 1 | - | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| **Denver Nuggets** | 1 | 1 | 1 | 0 | 0 | 0 | - | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| **Golden State Warriors** | 0 | 1 | 0 | 0 | 1 | 0 | 1 | - | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| **Houston Rockets** | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | - | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| **Los Angeles Lakers** | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | - | 0 | 0 | 1 | 0 | 1 | 1 |
| **Miami Heat** | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | - | 0 | 1 | 1 | 1 | 0 |
| **Milwaukee Bucks** | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | - | 1 | 1 | 1 | 0 |
| **New York Knicks** | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | - | 0 | 0 | 1 |
| **Philadelphia 76ers** | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | - | 1 | 1 |
| **Phoenix Suns** | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | - | 0 |
| **Toronto Raptors** | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | - |

## (c) Away Matchup Counts: $c^A(i, j)$

Each cell shows the number of times the row team (*i*) played away at the column team's (*j*) arena.

| Team \ Opp. | ATL | BOS | BKN | CHI | CLE | DAL | DEN | GSW | HOU | LAL | MIA | MIL | NYK | PHI | PHX | TOR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Atlanta Hawks** | - | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| **Boston Celtics** | 1 | - | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| **Brooklyn Nets** | 0 | 0 | - | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| **Chicago Bulls** | 1 | 0 | 0 | - | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 |
| **Cleveland Cavaliers** | 0 | 0 | 0 | 0 | - | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| **Dallas Mavericks** | 1 | 1 | 1 | 0 | 0 | - | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 0 |
| **Denver Nuggets** | 0 | 0 | 1 | 1 | 1 | 1 | - | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| **Golden State Warriors** | 1 | 1 | 1 | 1 | 0 | 1 | 0 | - | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |

| Team \ Opp. | ATL | BOS | BKN | CHI | CLE | DAL | DEN | GSW | HOU | LAL | MIA | MIL | NYK | PHI | PHX | TOR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Houston Rockets** | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | - | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| **Los Angeles Lakers** | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | - | 1 | 1 | 1 | 1 | 0 | 0 |
| **Miami Heat** | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | - | 1 | 0 | 0 | 0 | 1 |
| **Milwaukee Bucks** | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | - | 0 | 0 | 0 | 1 |
| **New York Knicks** | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | - | 1 | 1 | 0 |
| **Philadelphia 76ers** | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | - | 1 | 0 |
| **Phoenix Suns** | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | - | 1 |
| **Toronto Raptors** | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | - |

# Question 2

We wrote an Integer Program whose feasible solutions are all the valid schedules that preserve all structural properties of the original schedule while potentially reassigning which teams play each other on each date.

## Sets and Indices

$T$ = set of all teams ($|T| = 16$)

$D$ = set of all game dates ($|D| = 16$)

$i, j \in T$ denote teams ($i \neq j$)

$d \in D$ denotes a game date

## Parameters

From Question 1:

$H(i)$ = set of dates on which team $i$ plays at home

$A(i)$ = set of dates on which team $i$ plays away

$c^H(i,j)$ = number of times team $i$ hosts team $j$

$c^A(i,j)$ = number of times team $i$ visits team $j$'s arena

## Decision Variables

Define binary variables:

$$x_{ijd} \in \{0, 1\} \quad \text{for all } i \neq j \in T,\ d \in D$$

where $x_{ijd} = 1$ if and only if team $i$ hosts team $j$ on date $d$.

## Constraints

### (e) Home Dates Preserved

For each team $i$ and date $d$, team $i$ plays exactly one home game if $d$ is a home date, and none otherwise:

$$\Sigma_{j \neq i}\ x_{ijd} = 1\ [d \in H(i)] \quad \text{for all } i \in T,\ d \in D$$

On each of team *i*'s original home dates, exactly one opponent visits, on all other dates no home game is played.

### (f) Away Dates Preserved

For each team *i* and date *d*, team *i* plays exactly one away game if *d* is an away date, and none otherwise:

$$\Sigma_{j \neq i}\ x_{jid} = 1\ [d \in A(i)] \quad \text{for all } i \in T,\ d \in D$$

On each of team *i*'s original away dates, team *i* visits exactly one opponent, on all other dates no away game is played.

### (g) Home Matchup Counts Preserved

Over all dates, team *i* hosts team *j* exactly $c^H(i,j)$ times:

$$\Sigma_{d \in D}\ x_{ijd} = c^H(i,j) \quad \text{for all } i \neq j \in T$$

The total number of home games team *i* plays against each specific opponent *j* is preserved from the original schedule.

### (h) Away Matchup Counts Preserved

Over all dates, team *i* visits team j's arena exactly $c^A(i,j)$ times:

$$\Sigma_{d \in D}\ x_{jid} = c^A(i,j) \quad \text{for all } i \neq j \in T$$

The total number of away games team *i* plays at team *j*'s arena is preserved from the original schedule.

## Complete Formulation

The full IP (without objective) is:

**Find** $x_{ijd} \in \{0,1\}$ for all $i \neq j \in T,\ d \in D$

**subject to:**

$\Sigma_{j \neq i}\ x_{ijd} = 1[d \in H(i)]$ for all $i \in T$, $d \in D$ (e)

$\Sigma_{j \neq i}\ x_{jid} = 1[d \in A(i)]$ for all $i \in T$, $d \in D$ (f)

$\Sigma_{d \in D}\ x_{ijd} = c^H(i,j)$ for all $i \neq j \in T$ (g)

$\Sigma_{d \in D}\ x_{jid} = c^A(i,j)$ for all $i \neq j \in T$ (h)

$x_{ijd} \in \{0,1\}$ for all $i \neq j \in T$, $d \in D$

## Model Size and Result

Variables: 16 × 15 × 16 = 3,840 binary variables

Constraints: 2 × 16 × 16 + 2 × 16 × 15 = 992 linear constraints

Running our model with Gurobi returned "Status: Optimal", confirming the constraint system is consistent and at least one valid reassignment of opponents exists.

# Question 3

## Additional Constraint

Building on the feasibility model of Question 2, we add the constraint that limits how many time zones a team must cross over any three consecutive games. For each team *t* and each triple of consecutive

game dates ($d_1$, $d_2$, $d_3$) in its schedule, the sum of consecutive time zone differences must not reach 4 or more:

$$|tz(t,d_2) - tz(t,d_1)| + |tz(t,d_3) - tz(t,d_2)| \leq 3 \quad \text{for all } t \in T, \text{ consecutive triples } (d_1,d_2,d_3)$$

where tz(t, d) denotes the UTC offset of the arena where team *t* plays on date *d*. Each arena is assigned to one of four time zone groups: Eastern (UTC−5), Central (UTC−6), Mountain (UTC−7), or Pacific (UTC−8).

## Linearization

Since tz(t, d) is itself a linear expression in the x variables (each team plays in exactly one arena per game date), the absolute-value terms are linearized in the standard way. Let $\delta_{12} = tz(t,d_2) - tz(t,d_1)$ and $\delta_{23} = tz(t,d_3) - tz(t,d_2)$. We introduce auxiliary variables $p_{12}, p_{23} \geq 0$ for each (team, triple) pair:

$$p_{12} \geq \delta_{12}, \quad p_{12} \geq -\delta_{12} \quad \Rightarrow \quad p_{12} = |\delta_{12}|$$

$$p_{23} \geq \delta_{23}, \quad p_{23} \geq -\delta_{23} \quad \Rightarrow \quad p_{23} = |\delta_{23}|$$

$$p_{12} + p_{23} \leq 3$$

## Result (Infeasible)

The model is infeasible. Gurobi proves that no schedule satisfying all constraints (e)-(h) from Question 2 can simultaneously satisfy the time zone constraint for every team.

To diagnose this, we computed the IIS (Irreducible Inconsistent Subsystem), the smallest subset of constraints that is still collectively infeasible. The IIS Gurobi returns is concentrated almost entirely on the Golden State Warriors: it contains 31 structural constraints from (e)-(h) along with 3 time zone constraints, all 3 of which involve Golden State on triples k = 2, 4, and 10 of its game sequence. An IIS is a minimal infeasible subset rather than a unique one (different solver runs could surface different minimal sets) but the result is intuitive. The time zone difference from Pacific (UTC−8) to Eastern (UTC−5) is 3 hours, so any back-to-back crossing of those boundaries produces a sum of 4 or more, and Golden State's home/away pattern forces several such crossings.

## Relaxation with Slack Variables

Since the hard constraint is infeasible, we relax it by introducing a non-negative slack variable *s(t, k)* for each team *t* and consecutive triple *k*. The constraint becomes:

$$p_{12} + p_{23} \leq 3 + s(t,k), \quad s(t,k) \geq 0$$

We then minimize the total slack across all teams and triples:

$$\min \ \Sigma_{t,k} \ s(t,k)$$

This finds the schedule that comes closest to satisfying the time zone constraint (the one that violates it by the least possible total amount). The optimal total slack is **27** hours, distributed across 9 teams and 19 triples. Golden State Warriors accounts for the largest single share (10 hours across 5 triples), followed by Phoenix Suns (4 hours, 3 triples) and the Lakers (3 hours, 3 triples). The remaining 6 teams contribute 1-2 hours each. By comparison, the original draft schedule has 22 violating triples and 33 hours of total excess, so the IP reduces unavoidable time zone burden by 6 hours (~18%) without changing any team's home/away dates or matchup counts. The 27-hour figure is also a lower bound: no schedule preserving (e)-(h) can achieve a smaller total violation. The corresponding schedule can be found in the file *schedule_q3_relaxed.csv*.

# Question 4

For Question 4, we considered two different ways to improve the current schedule. Method 1 minimizes the total distance travelled across all teams. Method 2 minimizes the maximum deviation from the league-average travel distance.

## Computing Travel Distance

For each method, travel distances are computed as Haversine distances in miles between consecutive arena locations for each team. For team *t* travelling between consecutive game dates $d_1$ and $d_2$, four cases arise depending on whether each game is home or away:

Home → Home (HH): zero travel (both games are at the team's own arena).

Home → Away (HA): dist(arena(t), arena($h_2$)) where $h_2$ is the host on $d_2$. This is linear in x.

Away → Home (AH): dist(arena($h_1$), arena(t)) where $h_1$ is the host on $d_1$. Also linear in x.

Away → Away (AA): dist(arena($h_1$), arena($h_2$)), which is a product of two binary variables (bilinear).

## Linearization of Away → Away Legs

The bilinear AA terms are linearized by introducing auxiliary binary variables $w(t, d_1, d_2, h_1, h_2) \in \{0, 1\}$ for each (team, date pair, host pair) combination:

$$w \leq x_{h_1,t,d_1}, \quad w \leq x_{h_2,t,d_2}, \quad w \geq x_{h_1,t,d_1} + x_{h_2,t,d_2} - 1$$

These three constraints enforce $w = x(h_1,t,d_1) \cdot x(h_2,t,d_2)$, the standard linearization of a binary product. The travel contribution of each AA leg is then dist(arena($h_1$), arena($h_2$)) · w, which is linear.

## Method 1 - Minimizing Total Travel Distance

$$\min \ \Sigma_{t \in T} \ \text{travel}(t) \quad \text{subject to constraints (e)-(h)}$$

This objective targets league-wide efficiency: the total miles flown across all 16 teams. Reducing the sum is a natural improvement metric for the league since it correlates directly with charter flight cost and time spent on the road. Because the original schedule is itself a feasible point of this IP, the optimum total can only be smaller or equal to it.

## Model Size

Variables: 3,840 x-vars + ~13,275 w-vars = ~17,115 binary variables

Constraints: 992 base + 3 × (number of AA legs) linearization constraints for w

## Results

The total travel leaguewide went from 200,087 miles in the original schedule to 192,082 miles, a 4% reduction. The maximum team travel under this objective is 20,517 miles (Golden State, essentially unchanged from the original 20,400). This is expected: with the total-travel objective, the model has no incentive to reduce any single team's travel beyond what helps the sum, so individual teams may go up or down. Toronto and Brooklyn each gain ~1,300 miles while Boston, Cleveland, and several others lose more than 1,000. This optimized schedule can be found in *schedule_min_total.csv*.

**Travel comparison (original vs. optimized):**

| Team | Original (mi) | Optimized (mi) | Change (mi) |
|---|---|---|---|
| **Golden State Warriors** | 20,400 | 20,517 | +117 |
| **Phoenix Suns** | 16,730 | 15,971 | -759 |

| Team | Original (mi) | Optimized (mi) | Change (mi) |
|---|---|---|---|
| **Los Angeles Lakers** | 16,166 | 15,199 | -967 |
| **Boston Celtics** | 15,756 | 13,520 | -2,236 |
| **New York Knicks** | 15,669 | 14,637 | -1,032 |
| **Philadelphia 76ers** | 13,977 | 12,864 | -1,113 |
| **Dallas Mavericks** | 13,573 | 13,311 | -262 |
| **Miami Heat** | 12,798 | 12,503 | -295 |
| **Brooklyn Nets** | 12,385 | 13,630 | +1,245 |
| **Toronto Raptors** | 10,789 | 12,134 | +1,345 |
| **Denver Nuggets** | 9,922 | 8,756 | -1,166 |
| **Cleveland Cavaliers** | 9,688 | 8,320 | -1,368 |
| **Houston Rockets** | 9,204 | 8,528 | -676 |
| **Milwaukee Bucks** | 8,458 | 8,258 | -200 |
| **Atlanta Hawks** | 7,308 | 7,234 | -74 |
| **Chicago Bulls** | 7,265 | 6,701 | -564 |
| **TOTAL** | **200,087** | **192,082** | **-8,005** |

## Method 2 - Minimizing Maximum Deviation from League-Average Travel

As an alternative improvement, we also solved a model that minimizes the maximum deviation of any team's total travel distance from the league-average travel distance. Let $travel(t)$ denote the total travel distance of team $t$, and let $\bar{D}$ denote the average travel distance across all teams. Introducing a scalar variable $\delta$, the objective is:

$$\text{Min}(\delta)$$

subject to:

$$\delta \geq travel(t) - \bar{D} \quad \forall t \in T$$
$$\delta \geq \bar{D} - travel(t) \quad \forall t \in T$$

where:

$$\bar{D} = \frac{1}{|T|} \sum_{t \in T} travel(t)$$

This is a min-max fairness formulation: $\delta$ measures the largest absolute deviation from the league-average travel, and minimizing it makes team travel as balanced as possible.

## Model Size

Variables: 3,840 x-vars + ~13,275 w-vars + 16 travel-vars + 1 $\delta$= ~17,132 binary/continuous variables

Constraints: 992 base + 16 travel-definition constraints + 32 fairness-deviation constraints + w linearization constraints

## Results

Under this fairness-based objective, the optimized schedule achieves a league-average travel distance of 13,140 miles and reduces the maximum deviation from the league average from 7,895 miles to 5,000 miles, which corresponds to a **36.7%** improvement in fairness. The travel range is also reduced from 13,135 miles to 9,882 miles. This optimized schedule can be found in *schedule_min_dev.csv*.

### Team-by-Team Travel Comparison: Original vs. Optimized Schedule

| Team | Original | Optimized | vs Orig Avg | vs Opt Avg |
|---|---|---|---|---|
| **Golden State Warriors** | 20,400 | 18,045 | +7,895 | +4,905 |
| **Phoenix Suns** | 16,730 | 15,532 | +4,225 | +2,392 |
| **Los Angeles Lakers** | 16,166 | 14,909 | +3,660 | +1,769 |
| **Boston Celtics** | 15,756 | 18,140 | +3,251 | +5,000 |
| **New York Knicks** | 15,669 | 17,681 | +3,163 | +4,542 |
| **Philadelphia 76ers** | 13,977 | 14,614 | +1,472 | +1,474 |
| **Dallas Mavericks** | 13,573 | 13,390 | +1,067 | +251 |
| **Miami Heat** | 12,798 | 12,986 | +292 | -154 |
| **Brooklyn Nets** | 12,385 | 13,881 | -120 | +741 |
| **Toronto Raptors** | 10,789 | 11,665 | -1,716 | -1,474 |
| **Denver Nuggets** | 9,922 | 9,376 | -2,583 | -3,764 |
| **Cleveland Cavaliers** | 9,688 | 13,727 | -2,817 | +587 |
| **Houston Rockets** | 9,204 | 11,274 | -3,302 | -1,866 |
| **Milwaukee Bucks** | 8,458 | 8,258 | -4,048 | -4,881 |
| **Atlanta Hawks** | 7,308 | 8,313 | -5,198 | -4,827 |
| **Chicago Bulls** | 7,265 | 8,443 | -5,241 | -4,697 |
| League average | 12,505 | 13,140 | | |

### Fairness Summary for the Original and Fairness-Optimized Schedules

| Metric | Original | Optimized | Change (mi) |
|---|---|---|---|
| **Max deviation from avg (δ)** | 7,895 | 5,000 | -2,895 |
| **Range (max - min travel)** | 13,135 | 9,882 | -3,253 |
| **Max travel** | 20,400 | 18,140 | -2,260 |
| **Min travel** | 7,265 | 8,258 | +993 |
| Total travel | 200,087 | 210,232 | +10,145 |

Method 1 and Method 2 improve the schedule according to different objectives. Method 1 minimizes total leaguewide travel, prioritizing efficiency: it cuts 8,005 miles (~4%) from the original 200,087-mile total, but with the Warriors' worst-case essentially unchanged (20,400 → 20,517) and several teams traveling more than they did before. Method 2 minimizes the maximum deviation from the league average, prioritizing fairness: it compresses the spread (max deviation 7,895 → 5,000 mi, range 13,135 → 9,882 mi) and brings Golden State's travel down by over 2,000 miles, but at a 5% increase in total leaguewide travel (200,087 → 210,232 mi). The two methods produce different schedules and help different teams, which highlights an efficiency-fairness tradeoff. Minimizing total miles traveled doesn't balance the load, and balancing the load will cost the league miles. Depending on the league’s main objective however, we believe one of these improvements could definitely help optimize the schedule.

## Conclusion

Across the four questions, integer programming gave us a clean and flexible way to reason about the schedule. The Question 2 model showed that preserving each team's home/away pattern and matchup counts still leaves room for opponents to be reassigned. Question 3 showed that adding a hard time zone constraint to that base model is infeasible: the IIS analysis pointed to Golden State as the team most exposed to coast-to-coast jumps, and a relaxation with slack variables proved that any schedule preserving the original structure must accept at least 27 hours of total time zone violation (a 6-hour or ~18% reduction from the original 33 hours). Question 4 showed that the league has real room to improve travel: minimizing total miles cuts 8,005 miles (~4%) leaguewide while leaving the worst-traveled team essentially unchanged, while minimizing the maximum deviation from the league average compresses the spread by 37% but adds about 5% to total mileage. The two methods produce genuinely different schedules and highlight a clear efficiency/fairness tradeoff.

The model has limitations worth noting. Our 16-team, 16-date instance is much smaller than a full 30-team, 82-game season, and the AA bilinear linearization scales quadratically in the number of arenas, so a real-world version would need stronger formulations or decomposition methods. We also hold the home/away date pattern fixed, while allowing it to vary would unlock further improvement but would change the problem the assignment specifies. Haversine distance approximates flight mileage but ignores actual flight routing and charter logistics, and our static UTC offsets ignore daylight saving. Natural extensions include adding back-to-back limits, rest-day constraints between games, and arena availability windows. Combining the efficiency and fairness objectives with a weighted sum would also let the league explicitly tradeoff between the two.

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